

S-DES Algorithm Guide

1. Reference Tables (Rules)

P10

Pos	1	2	3	4	5	6	7	8	9	10
Val	3	5	2	7	4	10	1	9	8	6

P8

Pos	1	2	3	4	5	6	7	8
Val	6	3	7	4	8	5	10	9

IP

Pos	1	2	3	4	5	6	7	8
Val	2	6	3	1	4	8	5	7

S-Boxes

S_0				S_1			
1	0	3	2	0	1	2	3
3	2	1	0	2	0	1	3
0	2	1	3	3	0	1	0
3	1	3	2	2	1	0	3

2. Key Generation Steps

Raw Key: 1010000010

Step 1	P10 Permutation ^{1 2 3 4 5 6 7 8 9 10} 1 0 0 0 0 0 1 1 0 0
Step 2	Split & Shift (LS-1) L: ^{1 2 3 4 5} 0 0 0 0 1 R: ^{1 2 3 4 5} 1 1 0 0 0
Step 3	Generate K1 (P8) ^{1 2 3 4 5 6 7 8} 1 0 1 0 0 1 0 0 ✓
Step 4	Shift Left 2 (LS-2) L: ^{1 2 3 4 5} 0 0 1 0 0 R: ^{1 2 3 4 5} 0 0 0 1 1
Step 5	Generate K2 (P8) ^{1 2 3 4 5 6 7 8} 0 1 0 0 0 0 1 1 ✓

3. Encryption Steps

Plaintext: 01110010

#	Stage	Process Visualization
1	IP	$\begin{array}{cccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \boxed{1} & \boxed{0} & \boxed{1} & \boxed{0} & \boxed{1} & \boxed{0} & \boxed{0} & \boxed{1} \end{array}$
ROUND 1 (Key K_1: 10100100)		
2	Func f_K	1. Expand Right $\rightarrow 11000011$ 2. XOR $K_1 \rightarrow 01100111$ 3. S-Boxes $\rightarrow 1011$ 4. P4 Perm $\rightarrow \begin{array}{cccc} 1 & 2 & 3 & 4 \\ \boxed{0} & \boxed{1} & \boxed{1} & \boxed{1} \end{array}$
3	XOR L	$L_0(1010) \oplus 0111 = \text{L1: } 1101$
SWAP (Left $\leftrightarrow$ Right)		
ROUND 2 (Key K_2: 01000011)		
4	Func f_K	1. Expand Right $\rightarrow 11101011$ 2. XOR $K_2 \rightarrow 10101000$ 3. S-Boxes $\rightarrow 1011$ 4. P4 Perm $\rightarrow \begin{array}{cccc} 1 & 2 & 3 & 4 \\ \boxed{0} & \boxed{1} & \boxed{1} & \boxed{1} \end{array}$
5	XOR L	$L(1001) \oplus 0111 = \text{L2: } 1110$
6	Final	Combine $L_2R_2 \rightarrow 11101101 \xrightarrow{IP^{-1}}$ $\begin{array}{cccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \boxed{0} & \boxed{1} & \boxed{1} & \boxed{1} & \boxed{0} & \boxed{1} & \boxed{1} & \boxed{1} \end{array}$

4. Decryption Steps

Ciphertext: 01110111

#	Stage	Process Visualization
1	IP	$\begin{array}{cccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \boxed{1} & \boxed{1} & \boxed{1} & \boxed{0} & \boxed{1} & \boxed{1} & \boxed{0} & \boxed{1} \end{array}$
ROUND 1 (Key K_2: 01000011)		
2	Func f_K	1. Expand Right $\rightarrow 11101011$ 2. XOR $K_2 \rightarrow 10101000$ 3. S-Boxes $\rightarrow 1011$ 4. P4 Perm $\rightarrow \begin{array}{cccc} 1 & 2 & 3 & 4 \\ \boxed{0} & \boxed{1} & \boxed{1} & \boxed{1} \end{array}$
3	XOR L	$L(1110) \oplus 0111 = \text{New R: } 1001$
SWAP (Left $\leftrightarrow$ Right)		
ROUND 2 (Key K_1: 10100100)		
4	Func f_K	1. Expand Right $\rightarrow 11000011$ 2. XOR $K_1 \rightarrow 01100111$ 3. S-Boxes $\rightarrow 1011$ 4. P4 Perm $\rightarrow \begin{array}{cccc} 1 & 2 & 3 & 4 \\ \boxed{0} & \boxed{1} & \boxed{1} & \boxed{1} \end{array}$
5	XOR L	$L(1101) \oplus 0111 = \text{Final L: } 1010$
6	Final	Combine $LR \rightarrow 10101001 \xrightarrow{IP^{-1}}$ $\begin{array}{cccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \boxed{0} & \boxed{1} & \boxed{1} & \boxed{1} & \boxed{0} & \boxed{0} & \boxed{1} & \boxed{0} \end{array}$